ActInf GuestStream: Ron Itelman: Cognitive Experience Design (CXD): Des

1:03:49

All right. Hello and welcome. It's January 17th, 2025. And we're in Active Inference

Guest Stream 96.1 with Ron Edelman discussing Designing Intelligent Experiences for Trust.

So thank you, Ron, for joining and looking forward to your presentation and the discussion.

All right. So really happy to be here. I must say that I'm only here because I randomly reached out to Carl and he was very kind and generous.

And he decided to, you know, entertain some conversations. And those, this has been about two years,

those conversations led to a couple of presentations, meeting some really incredible people involved in Active Inference.

And it really kind of set me on a new course, new ways of thinking.

And I wanted to share a more practical, perhaps implementation practitioner approach to what are the real world problems we're seeing

with integrating the brilliant community of people who are trained in Bayesian statistics.

It's a completely different way to think about problem solving.

And if you start working with product managers, with designers, business folks,

they might have just a very different framework for thinking about how to create systems.

And I just want to kind of share an experience I had, which kind of changed my life.

And it was about music.

And I was really struggling to learn the Western style of music.

And I had a teacher who studied in India.

And so they have a different way.

It's instead of writing things down in straight lines and you, you sing your music first.

It's called conical.

So quarter notes would be takidimi, takidimi, takidimi, takidimi, takidimi, right?

Triplets would be takita, takita, takita, takita, takita.

And that system of learning accelerated me more than in a few weeks than years of like taking music lessons.

And the reason that this I think is important is because it introduced me to the idea around

having a trusted source of truth in collaboration.

All of a sudden, when I was playing music, we could all agree, you know, metronomes, timing,

it opened up an entire language for me for how to interface with others, be completely creative,

have that freedom to collaborate.

But it was also safe, right?

Like it was also we could remain within the constraints of the system.

And that unlocked all kinds of incredible learning experiences for me.

And since then, I've viewed information flowing through organizational networks, kind of through

the lens of music.
Like what is our metronome that allows us to collaborate when we're building software, when we're providing recommendations to customers, when we're troubleshooting operational and business processes.
And I think Carl enjoyed my analogy of synchronization and music when talking about these systems.
So he called it computational psychiatry, not words I would use.
But that's, that's how this journey kind of began for me.
So just a little bit about me.
I've written two books now.
The first one was came out last year.
It's unifying business data and code.
It was, it was incredible to write an O'Reilly book.
And that book deal got me a second book deal, The Language of Innovation, which is coming out in a couple of months.
And my life has been around meeting like incredibly talented people and figuring how do we leverage that creative brain power and passion and help organizations minimize their costs.
And I think that I view myself as kind of a bridge, a connector between all these different worlds.
And unifying business data and code was kind of my first exploration into how we formalize methodologies to get everyone to have, quote unquote, a metronome.
So how did we get to this, to the idea of like, why is trust interesting?
Well, it started with a conversation with Carl, I think at his theoretical neurobiology group, where I asked, what is the definition of understanding?
And he was like, well, it's quite simply, it's the, it's a shared model, right?
If you both have the same inputs, you should both have the same outputs.
And that for me was how I connected back to, to the idea of the metronome.
Because in reality, we don't have shared metronomes in a business.
In fact, I would say, if you go to two different departments, and you say, what does this word mean?
Usually they will give me a different definition.
And even if I ask two people within the same department, just as a pure consulting kind of approach on process optimization or architecture, explain to me how this process works, it is almost never described to me the same.
So this is a really hard problem, because if we're going to be able to have, you know, AI make decisions, and people can't even agree on what the operational reality is, how is an AI going to do that?
So the metronome for me is just a symbol of, it's always something I could trust.
It was just something I could like, let go of holding on to what I think is right.

And I just, this is the truth, and everyone else can align around it.
And it was really interesting for me to approach the problem with like, how would you computationally define it?
And I will tell you, I do not have a computational definition of trust right now.
But I am thinking about frameworks, approaches, ways to quantify and engineer trust in systems.
And this to me is a very, very exciting problem.
So the best analogy I had was when I was just, I was feeling a little tired, and I started, you know, watching, watching some TV, and I pull up the movie Edge of Tomorrow.
Now, for those of you that don't know Edge of Tomorrow, it's like, it's Groundhog Day meets sci-fi.
Tom Cruise keeps dying over and over in this battle with aliens.
And each time he wakes up, he's learned from his previous experiences.
But there's one scene in the movie, which really kind of helped me think of ways to articulate ways we can think about trust.
So in the beginning, Tom's like smiling, he's laughing, you know, 100% feeling safety and control.
The general says, you're going to go out into the battlefield.
And he's a, Tom Cruise is a reporter, you're going to report on the, on the rule, on the war.
Tom's afraid of this.
And so he threatens the general, if you send me into battle, I'm going to basically make you look bad.
And so then the general then says, I'm going to call you a deserter.
And he throws Tom, not into a reporter role, but actually into this, you know, mechanized combat suit.
And Tom doesn't even know how to use the suit.
And eventually the, the sergeant says, jump or die.
And it's in this particular moment where it's like the opposite of trust kind of felt clear to me, right?
He has no certainty of what's going to happen.
It's pure chaos.
He has no control over these events.
He has no safety because he's, he's going to die possibly.
And, you know, he has no understanding of like, what the heck is even happening to me.
And, and if we look back at how LLMs and not everything is LLMs, but LLMs seem to be very hot.
And, but I think this is true in, in, for any kind of machine learning probabilistic system.
When we are in situations, granted, it's not the edge of tomorrow, that situations, but when

we are in a situation where we have to make decisions that could be life or death, it could
be patient care.
It could be the life or death of a company.
If you make a wrong decision, it could be the difference between layoffs, right?
When we use these probabilistic systems and we're leaving the deterministic approach, we
have a class because there's a lot of attention on, you know, fine tuning and prompt engineering
and, you know, trying to force these AI systems to behave deterministically, but they are not
deterministic.
And so I want to, I want to think about these four dimensions around certainty, control, safety,
and understanding, right?
So when we have what's, if you're familiar with the paper called tool former, it introduces
these kind of two terms, post-processing and pre-processing.
And so what OpenAI did when they first kind of shaped the entire industry, right?
That we, that we now are thinking about, it's you give us all of your, you know, put everything
in a vector database that you send, you do your fine tuning, your prompt engineering, and,
you know, our AI is going to be super, you know, quote unquote, super intelligence, general
intelligence.
It's really good if that's your business model to have companies do that, right?
It's in their benefit.
It's in the cloud provider's benefit for a business to have really messy data.
And of course, pre-processing is hard because you can't possibly know in advance all the ways
things will be used and all the rules.
And we're left with this kind of black box situation.
And it leaves this black box situation obviously is very ambiguous, right?
And when we're going shopping and we want to buy a cup of coffee, we don't want to have
a probabilistic answer to what a cup of coffee costs.
We want to know this is how much it costs every single time when we're going to make our
decision, right?
And so how do we bring in these probabilistic and deterministic systems into user experiences?
Another dimension around safety, compliance is typically governance kind of things is just
sharing my experience.
First of all, if you're into, if you're a taxonomist or, you know, if your knowledge graphs
and linked data are your thing, incredibly powerful, great.
But I do want to share with people what the problems I see from in the real world, companies
that are trying to implement these approaches.
You have a very small group of people who cannot possibly know everything in the company
and every operation and define things.
So they become quickly become bottlenecks for these top down rules.

And the bottom up requests become an overwhelming tsunami.

So people just go around it.

And now you have the worst of these worlds, right?

Because you're paying for enterprise software that you're also not using.

And you have all this like shadow, you know, shadow IT kind of things going on.

And when we, you know, thinking of biology, like when ants leave these trails,

we typically have a really hard time leaving trails of how we solve tasks.

But if we can find a way to leave trails, these audit trails of how do we solve cognitive tasks,

we create the opportunity for kind of an organizational scale learning, right?

So the difference between the top down approach and this complex adaptive systems approach

is if you just can observe how people are solving tasks,

you will, you will have a fluid way to see what are the rules and processes people are following in the real world.

Autonomously with freedom, et cetera, et cetera.

And how did those, how, how are the interactions with governance?

And this is, this is what was the origin of what I was really interested in working on.

So I had a previous startup where it, we basically pulled data, all public data from GitHub.

If you're a developer and you did some code, we had data on you and your code and you're done to the commit level.

And we were trying to sell it to recruiters like, hey, do you want developers?

We can analyze people's codes and recommend developers for you to recruit.

But I had a problem.

I had two weeks for a big demo to like, you know, one of these fortune 50 companies.

And we are, all our data was in a SQL database.

And I wanted to use AI and I didn't have time to learn Langchain and, you know, figure out how to convert things into a vector database.

I had to hack something together, which would enable me to use natural language to query a database.

And so I decided to take a really old school approach, just using regular expressions.

And that is what I'm trying to formalize into a protocol, which I call Trust Loop.

So before I go into anything, Daniel, are there any questions coming up or any comments or should I proceed?

Not yet.

Proceed.

Thanks.

Very good.

So we want to take these kind of four dimensions and just think about them in terms of how we're designing systems where people can learn and then the whole organization can learn.

And where I think this could be really interesting is, you know, someone responded as I started posting about this, about the EU AI Act.

How are you going to be compliant?

What's the picture look like?

First of all, we need to integrate into a user interface.

And then we need to be able to have a registry of rules.

The approach I took is to have an LLM look at my rules and then recommend suggestions to dynamically improve.

With a feedback loop, any kind of rule management for me as an individual, but it could also be for a system if there are like compliance or regulatory requirements.

And then lastly is this kind of cognitive audit trail, right?

So I'm going to give you a very super simple example.

But let's say I, and this is literally the most simple example I could come up with.

But let's say, for example, I have this chat interface and I want to ask a question about my data, right?

Like let's say I have some data set or database and I want to calculate revenue.

So like if I were to post a question, you know, can you tell me what my revenue was, et cetera, et cetera.

And there are clearly some bugs.

So I'm just simply asking up here a feedback loop.

What do you mean by revenue?

Do you mean net or gross?

Because these are wildly different calculations.

This is incredibly, incredibly simple.

It is merely identifying possible ambiguity in my language and then saying, hey, is this what you mean?

Or, hey, do you want to use the revenue calculation from this particular database?

Now, why is this important?

Well, let's go back to this slide here.

Because I think the important question is why use AI at all?

If you can accomplish your task in the most efficient manner, that doesn't mean there needs to be AI involved.

So there was a really interesting metric from Versus recently, which I know everyone here is a fan of,

that it beat OpenAI's mastermind gameplay versus O1, I think the O1 model.

And it said O1, I think it was like, took five hours and cost $200 and was only 70% accurate.

And Versus was 100% accurate, took five seconds and like cost 50 cents.

That's amazing.

I love that.

I love that.

And this is no disrespect to Versus or anyone in AI.

But my immediate question was, hang on a second.

I'm pretty sure the algorithm to solve mastermind exists in a JavaScript file somewhere on GitHub.

And that means zero cost.

And instead of taking seconds, it's milliseconds.

And 100% accurate.

Actually, I checked it was Python code is the code I checked, but same thing, right?

And so it just kind of reinforced this idea around, hey, the whole industry is barreling at 100 miles an hour towards this post-processing paradigm.

And so it's not just a lot of time.

But why?

But why?

Why should we send everything through AI?

How do we know what we should send through AI and what we should not send through AI?

And so that is what the Trust Lead Protocol is meant to do, is we're trying to figure out how do we route which types of cognitive tasks already have rules and deterministic processes to solve,

and which ones don't exist in our rules registry, so that that is where AI can shine, right?

Because if the reality is, as I've read, I haven't personally audited, that a single query for kind of an enterprise, mid-enterprise company can cost a couple hundred dollars to process and hallucinates,

is that really cost-efficient, especially if there's a deterministic solution out there that works 100%?

And just to be clear on as much as I think AI is amazing at certain tasks, this just came across this morning, so I just want to put it up here.

I didn't believe it when someone posted this on Blue Sky, so I typed it in myself.

It's not the exact same wording, but I got the exact same result.

So then I'm like, okay, well, is this a Google thing?

Or maybe it's like, what if I give it to Claude?

So I gave this to Claude and I asked, you know, is this correct?

And Claude was like, yeah, no, your glass of water will not freeze at 27 degrees.

So I said, but isn't 27 degrees lower than 32 degrees?

And it said, oh, yes, expertise, as far less infrastructure, think that, like, slapping on AI is going to be safe and reliable.

So this is kind of a skill skill, which leads me to another question.

How do you even measure that?

so if you look at chatbot arena right I have the name right where you have two different

LLM models and people are picking what they prefer okay a model has a preferred answer

but um different people in different contexts with different language might have what they

call a preferred you know or correct answer and a preference is not necessarily a deterministic

correct answer

okay I'm gonna um pause there Daniel any any questions or should I continue I I'm writing

some things down a lot of cool points to discuss and if anyone watching live wants to write questions

if they can go for it continue though okay yeah I want this to be interactive and not me just

rambling on so uh if you have anything anything you're curious about uh or not so um what we're

hoping to do is to create a very basic protocol in my my meet protocol is a guarantee
if you're going to build these types of systems just a simple scaffolding of here or here is how
you do it and so what we need to do is have our intents predicted from our words
we need to reflect them back to the user so that we can see the intent of the system
um and I don't know if uh Riddhi's here but she's the one that kind of like uh she's amazing and she's
the one that kind of um introduced me to that idea around how do we see into the mind of our AI systems
like what is the belief of the system and I think if people are concerned about the EU AI Act that's
exactly what they want to be able to to have is how much uncertainty exists in your model
uh and in your system right because going back to that chatbot arena you might have preferences of
model answers but that is very different than how efficient the system is at completing a task
right like your model might have a good answer but is it what the person who's using the model
needs to solve their problem which is which is a much different bench
um especially when you start calculating the cost of running some of these models
um secondly there's security issues um but being able to configure rules for these intents
what is deterministic and um what is probabilistic right
um and um there's one thing that kind of shocked me oops there's one thing that kind of shocked me
that um because I'm in presenter mode I can't move things around uh but I don't know if you see here
using Claude the API I ran um prep my mouse wheels accidentally getting hit all the time okay so if running
Claude there's a um like 100 success rate if I ran it through the user interface but if I run it through
the API and this is a benchmark of math word problems that open AI released
the accuracy dropped to 30 percent
the user interface when you use the um the web app has all kinds of fine-tuning that you and I have no idea
exist it's a black box to us and when we run it through an API we don't have the benefit of that
um and the reason this is important is because when I um and I'm going to stop sharing my screen now
when I told my friend this who uses Claude on a regular basis for data engineering tasks
he had no idea so he had no idea he had to do all this extra because he's like in the web UI it works
once I get it working in the web UI why don't I just like you know automate it through APIs
not necessarily bother bother even bothering to benchmark it because he trusted the web user
interface so I think that we're really an immature kind of industry um trying to apply AI to solve
really hard problems that often have really significant safety compliance uh decision making
um impacts and I I don't think we really have a a way to understand how to design and engineer trust uh and
so that's that's what I'm passionate about I just kind of wanted to you know share with the community
some some of my research and you know I think there's there's I could ramble on here but I also
don't know if uh I'd like to like come take a pause and see if anyone has any questions or comments
awesome I will read a question from the live chat anyone else can write and then I can make some other
comments and questions okay reconfigurability trainer writes Ron did your work building the
architecture include non-compliance related performance KPIs if so did anything about that exercise surprise
you non-compliant paced performance other KPIs not related to system compliance got it so the if you

haven't seen the the startup unit unified AI um I was really surprised like reading all the logos that use their service um and I think they're they're really well supported by the hugging face uh leadership um they distilled it down to three metrics you know cost speed accuracy and you can test any model and so I haven't used their service but they seem to have a really um good way for you know as simple as possible as simple API calls for people that test various models various prompt engineering various fine tuning um for benchmarking that it has nothing to do with compliance I think though there's a disconnect for me around um I've worked with some brilliant data scientists and AI engineers who you know working on their local host local machine would have incredible results and then they good luck getting that into production uh in the real world with like you know uh outside of training uh I forget the term but when you when you have examples that are outside of your training set um and so that that just brings me back to this idea of benchmarking is absolutely important for models kind of you know will not get me to say anything else but like if you're not benchmarking in the real world how those models are interacting operationally and what the impact to the businesses um you know eventually someone in the finance department is going to is going to be like hey we we just invested five million dollars into AI and like uh we have to now have three million dollars worth of extra HR costs to maintain all these complex new systems and you know our sales or whatever our marketing we don't have any real improvements um I've been around to see those kinds of layoffs they're not they're not fun and so I think that the discipline that the business discipline um that a data scientist could benefit from is is really starting to include um thinking about benchmarks beyond simply you know model performance or something like that that's that's very interesting like to kind of echo a piece back there within a compliance testing sandbox skeleton or scaffold then other kinds of measures and performances can be evaluated but to pursue those measures only when the the vector of the performance isn't necessarily aligned with that of the compliance then is going to lead to some kind of go no-go with compliance down the road that might not be trivial but if it was trivial then it would have been trivial in any order but in situations where it isn't trivial then you want to make sure that you have the expressivity to test different compliance environments almost before you test like the size of the data for example but you you brought up a really important um point um so I'm not sure if people are familiar with the whole world models kind of buzz young Lee khan if I'm pronouncing his name right uh you know he's been tweeting of our posting you know LinkedIn about it and um um um Fei Fei is it Fei Fei Ling I'm not sure if I'm pronouncing his name right apologies um at Stanford she just raised 200 million dollars for her startup on world models um and there were a couple of things there that I just wanted to point out um the first one is when we have a simulator like a physics simulator right like um a car driving simulator and we can run that simulation a million times and then out of that million times maybe we can do some like you know deep learning on it that is a fixed rule system right there are there are rules just like when we're doing compliance our simulators have things they are physically bound to follow the rules are known just like when you have uh Alpha Zero playing chess or go whatever it is games are fixed rule systems in the real world in business the rules are constantly changing legislation changes the ways people operate

changes HR rules change language changes there's no fixed simulator you can have for a business
and so the the approach that we're proposing with like this the with trust the protocol is to is to say
okay if we cannot impose some top-down fixed rule stimulator but we can create those you remember I had
that image of the ants trails that that if we can collect all these like ant trails for how things are
actually getting done what do people mean by these uh you know different words you can you can
have a and I don't know the actual technical term for this but it's like a liquid simulator food
simulator it's it's a it's revealing things that are happening by these quasi rules that are emergent
and that hopefully can create world models um but not through a fixed system right that was the first
thing I wanted to say before I go on to the second thing the answer is that okay um the second thing
was I was lucky enough to do this like little case study for um precision medicine department at Stanford um
and they were doing a hackathon
um around people who had rare diseases so the precision medicine um lab had very few patients
but they had lots and lots and lots of data I mean genomic data proteomic data metabolomic data like
you name it they were just barraging them with tests to create the most massive data sets they could
um on the few people they had with these very rare diseases
and um to prepare for that hackathon I met with another Stanford team that was trying to create like
the largest physics data set in in the world uh I'm wondering if they were actually part of this
fei fei uh uh laying uh uh world model thing but what was really interesting that they did
is they collected all this data and then they ambiguated everything to numbers that you couldn't
see what the columns were of the data and I thought that was a really interesting approach and so I asked
the organizer of the hackathon what you know what was behind it and he goes because we found when we
have these data sets with the words in the columns the people who are experts right the trained PhDs in
their field regularly underperformed people who had no knowledge or expertise of that field but they
were purely machine learning experts and I thought that was um I thought that was very interesting as
well that doesn't tie into the to the um compliance thing but just this this idea around um creating
simulators and removing even our own language and constraints and how we think um and and just
completely you know trusting uh good practices uh those seem to be the highest benchmark
um um approaches at least talking to you know an expert in the field that created this massive physics
dataset yeah in the design architecture you outlined and kind of connecting it to let's say um employee
nestmate ants leaving their traces through a sandbox or something like that it made me think about
the metacognitive environment that cognitive tasks are performed in like if the company were playing
board games which model or method should be applied to each of the cognitive games yeah and then the
metacognitive task you could bring out you know the heavy or expensive one too often or you could bring
out the lightweight less performant one or narrow one too often so that's like sort of thinking too much
or too little these kinds of metacognitive or attentional tasks and it's almost like okay over
the last years we've started to see the upswing of the capacity of the sort of like horsepower of the
statistical probabilistic which has always existed as this dual thread along the more expert rule-based
regular expression type systems and then it's like okay so we know at the very least we have whether

we think about it with system one system two or you know thinking symbolically and rule-based or thinking probabilistically and and numerically and then it's like now where and how within an enterprise auditing and and compliance environment that also creates not just the required operations that are that are needed

needed but also the training data where it's like what are the low-hanging fruit that we can use the shorter

ladder for what are the high-hanging fruit that like no one has been able to hit what so what's our reach goal

it's like you know because we're starting to see we have all these different affordances to deploy that have

wildly different capacities and different costs you could outsource it to someone for thousands of dollars over weeks or we could do this for hundreds of dollars over minutes and when you burn

through hundreds in just hitting one query those are critical and they really add up yeah and that's i i kind of smiling because i was just like you got it right like that's exactly what we're trying to architect here is when do you use ai when not which model which approach when do you use an expert system um

um it's a really interesting problem because if you're measuring against operational performance

versus the model performance

you might have very simple hacks

which are way cheaper to implement faster to implement you know less it complexity and you know and uh i i feel like if that is completely off the radar right now right now because everyone's like rushing towards this new paradigm and you know it's not going to be a

yeah not every company is a frontier foundation model ai research lab so why use something more complex um i'll read another question from the chat and then anyone else can write more okay

andrew pesheya wrote nice talk all relevant points topics on trust regarding agent activity do you have an opinion on the uses of actor critic frameworks in the case of llms for output refinement fact checking etc

yeah great question um i think i heard did he was the word expert in there or opinion i'm not sure what it

was let me just say i just have an opinion i'm just i'm just going through you know it's funny because um

writing a book you're getting published now you get questions oh like i have some secret answer no i'm figuring all this out

just like everyone else um and i can tell you what i've been um interested in and where i think being

an author is awesome and where it sucks so being where being an author is awesome is that you're completely

removed from having to do any of the work right like i got to spend two years not coding and just

thinking about all these ideas and talking to the most interesting people i could and you know

what about this what about this right and so when we talk about these agentic um actor paradigms i can

share with you what i've learned but it's not from a practitioner point of view it is purely from

reading thinking you know imagining um and working on my own little kind of agentic uh little experiments

um so it's clearly very very valuable i think for me personally just even like through the chat ui

for me to write something and then you know claude will be like that's awesome that's great and then

you build it up and then like i'll i'll send it in a new chat and i'll say okay now you're going to be

the role of a critic tell me all the things that's that's wrong right go through that process and andrew

eng actually a friend uh a good friend um yoshi he he pointed me to a recent injury um talk where he

talks about the this emergence of ai product managers and he talks about this you know multi

actor role like you have the the first brainstorm partner then the critic and then the you know maybe
the engineer who has to take the critic and the brain and figure out how to build it you know and
and you can see now in the like in the lm chain of thought reasoning approaches they are definitely
you know they're even saying like hey criticizing or reviewing etc so it's clearly a pattern that
i think has great value and is being engineered and implemented um however
i i recently had an experience with someone from the active inference institute i don't know if lance is
here he's amazing i love him uh uh so incredibly helpful um i i recently worked on this paper
it's the first time i've ever submitted anything for a you know talk um rldm in ireland right
reinforcement learning multi uh disciplinary um conference on cognitive world models which is like
how do we take the world model approach introduced in 2018 where you create a simulation of a simulation
super cool um and now if we had trust loop protocol instead of a fixed rule simulator we take this like
emergent kind of data and i tried to write the paper using kind of this actor approach right i did my best
and then i was like hey before i submit to this conference i've never written an academic paper before
don't have a phd totally intimidated help me great okay then i said please critique it great you know
how to prove it and and i tried to go through that multi-actor approach to using llm which again i think
is incredibly valuable blah blah blah um and then i sent it to lance right someone who actually does
uh who i trust deeply right there's no better person other than carl i can think of right um and maybe
ready right so i he took a look at the paper and immediately he's like it's missing boom boom boom
he's like you should try asking chat gpt for help not knowing that i was using cloud like the whole time
and like it was really interesting for me to see that despite using that multi-actor approach
it none of the things that lance pointed out did i get feedback from those multi-actors
um and i think that's how do you benchmark that right yeah that's that's um awesome one one thought and
i've seen this also come up many many times is like even when a character or a persona prompt layer
is added so it's like okay you know you're an engineer or you're a product designer so so it's kind
of preempting trying to warm up or put these uh general purpose models into character in some sense
the response that that gets pinged back is always going to have been like washed or brought back towards
the center of gravity of this model that was trained on prior text and so sometimes it's like
if somebody has three authentic ideas one two three and then they ask for it to be upgraded and it looks
on a first pass like the professionalism has been brought up greatly but at this point we all know that
we can get those kinds of responses looking professional back rapidly and maybe it converges to
common idea three four five and if they had just sent the authentic one two three with the typos and all
it's like we know that it could have we could have added more and then also um seeing what isn't there
is very different than just augmenting what is there um i'll read a question from the chat on this
theme a little bit okay reconfigurability trainer wrote what are your expectations on how legacy
enterprise architecture will have to change to adapt to ai how much i.t change do firms have control over
versus locked in by service providers yeah i think this is the cr is like a critical critical point of what
i'm interested in with this presentation today is
there's okay there's two layers to this the first is a ton of working decades long investment in

infrastructure already exists so the idea that you're going to take decades worth of pdfs and data and dump it all into a vector database and do your prompt engineering and fine tuning and use that instead of all these systems and investment that work that's just insane to me i'm sorry like um until we can show that google can know that 27 degrees is below 32 degrees right like these are not trusted systems they're probabilistic systems systems and the the amount of cost it's just you know outstanding uh out astounding sorry numbers and so the idea and I know it's like people want to learn these skills you get a big K jump if you have like Langchain and VectorDB and you're you know and everyone's starting some scratch so you can become an expert quickly um but there's already a ton of infrastructure that works and so rather than fire hosing everything through AI um what's how do you control what is AI and what is not like how does even the business team decide what specific data is going to be good and useful and have an ROI to use that right it just it just feels like um uh it feels like businesses feel so much pressure to adopt this that the C level has so much pressure to show to shareholders here's how we are going to you know have competitive edge and we're going to be a leader in adopt AI they were like they haven't like first really thought through some of these questions and so I think that exist leveraging existing infrastructure is the number one thing companies can do is just like avoid AI when possible right but there are things that I can do that deterministic systems can't which leads me to the other side of the coin here which is I've been in organizations that have massive I mean it's like a million dollar a year in like these data catalog and you know data management systems and like you talk to the business leaders or the BI leaders and and the data teams and like they they're not even understanding each other right the business people say oh do this it seems so basic and then the data teams they do all this stuff and it's not matching expectations and you know the data teams are like we we're not mind readers for what the business wants right and so the problem there is that the existing infrastructure doesn't work that's where I think AI can shine that's where I think AI actually has an incredible advantage around when you have all this inefficiency in getting people to collaborate on IT infrastructure which is deterministic that's where I think an LLM or something can unify people on language and approaches in intent right so so I think that's it's a really important question that you just highlighted I think these are some key issues that that will that are that are to live with not to defeat like if we equate determinism with safety and probabilistic with danger then we're not even gonna get out of bed in the morning like in cognitive security we have one joke that the safest airport is a parking lot you're not gonna have any issues with airplanes there but the the surfaces and the processes that are being actually done they they can't be deterministically known ahead of time right little motifs can and so obviously again low-hanging fruit but then understanding like everything is a a threshold of tolerance there's always a probabilistic um elements to safety in the real soft and fuzzy world and so understanding how all of these difference um on one hand like constraints and strictures and formal language like what is reasonable and all of this what is risky and then on the other

hand that the the experiences in the actual like what cognitive science is pointing at yes people talk about affordances and umbels and all these different aspects that help us study it across different systems but like just taking it to people's day-to-day experiences and what is um how do we even accurately evaluate whether something is too risky or not or what these trade-offs are and um it's hard to have situational awareness it i'm i'm sure even without knowing at the enterprise scale because it is like mapping intents preferences what capacities and all these different features again the kinds of things that in cognitive science and active inference we work on with the models those are features in the world too and how do we even know like where when how what if and it seems like now is the time to be exploring in those directions yeah yeah and and we're almost at a at a time so i just want to throw out there i want i want to comment on that in a second but just before i forget um thank you all for anyone uh who's who's uh shared your time with me today um i definitely am looking to connect with people in the active inference community whether it's just to say hello if i want to meet people i want to learn or if you're interested and we're looking for a case study partner um to basically try out the trust loop protocol to this talk has now led to some conversations with investors so if you want to connect please connect because i'm i'm really excited about this these problems um and i would like to kind of leave on a on a note um around systems and um what you alluded to like this real world emergent kind of um situation we have in enterprises um there was a santa fe institute talk i don't know why i like ants i like ants so um so they were talking about um the efficiencies of like colonies and how well they can adapt and learn and solve problems and the the one thing i just it struck me that this the scientists entire research came down to like this one line and she said the density of feedback loops in a system um directly correlate to how quickly it can adapt and learn and learn and so the reason that this protocol is called the trust loop protocol is because it's really a ux kind of interface um via just an api to get that feedback across various systems between probabilistic and deterministic and blend them together and um yeah i don't know is there anything anything else daniel or uh is that how do you feel about i could make many comments on on ants i'm i'm um what comes to mind is um over the decades of ants research um people are trying to understand like how do on every continent except of course ironically antarctica how did these collective behavioral strategies work in all these different niches with different regularities so those are kind of like different business operating environments or like market conditions or something like that and so and there can be multiple species different strategies in the same niche so people have tried to apply all kinds of different approaches and famously eo wilson promoted the cast ergonomic hypothesis the idea that something like an adam smith 1776 type story like free market in the colony colonies that are better from this economic perspective are maximizing the amount of income food you know and outcome reproductively per unit of biomass so that that's one view um and that relates like optimal foraging a lot of other concepts and then as a contrasting voice to that sort of economic optimization researchers like my phd advisor
deborah gordon highlighted some things like you mentioned such as the the rate and the frequency of interactions the types of interactions and learning and developmental trajectories those capacities and those

patterns can dominate such that whatever the economics are they sort of are however they end up but they're
being driven by these micro level interactions and so it's like i when thinking about businesses or other
epistemic environments um just how two people can look at the same colony you know same institute same
company whatever it happens to be and both its viability economically can be seen from that like macroeconomic
perspective otherwise it's not going to persist but also the the micro interactions are the i mean if if um
if there's a great foraging plan but the pheromone evaporates too rapidly then it's not going to work
and so it's like where are we at with digital cognitive or knowledge work or information spaces
which is largely what we're talking about including how those are propagating into physical spaces logistics
and supply and all these different areas physical spaces that have the information space
in influencing them and then what are going to be the sort of macro structural properties of the system
that no nestmate could could grapple with and then how is that going to relate to the first person
considerations like accessibility felt sense of trust cognitive overhead
and and and and kind of like if those can connect then there will be at least local harmony and success
but there's way more ways for those not to connect way more unsuccessful collective behavioral strategies than
successful and so in the digital space where there's almost total degrees of freedom of design
what does that interface look like right and and and you're approaching it and also what is that
it's like interfaces all the way down market markup blankets all the way down what is that interface
between the business intelligence and the api call like all these different connectors but if if even one
one critical one it is not how it needs to be or not as functional as it could be
that's what improvement should seek to target and what proactive design should should seek to prevent
hopefully we'll see this is also new but i'm hoping if we can get a good case study
then we should be able to quantify quantify um some of these benchmarks around connectivity through
interfaces through different systems across tasks various models between deterministic and probabilistic
rules like that's that's my next step i hope we can come up with some empirical evidence um
what what zone would the case study be and then how can people contact you
what's the case study um uh and the the zone i don't know like talking to the head of the hospital for
one possible case study um a couple of incubators uh food industry i i don't know like there's all kinds
of conversations until someone actually signs an agreement right who knows nothing may come out of
it but i do think we have a compelling approach i think this is a really hard expensive problem and if
we can find someone who's willing to give us a chance just to like demonstrate as as you know i guess
you know i understand it's an n of one right but if we can demonstrate um that we can show we can show
the i want to use your language right the act itself of making these connections across various interfaces
just doing that can have value right it's the digital equivalent of pheromones right but that's that's
effectively what we're calling these cognitive auditable cognitive trails it's just meant to say
like we're leaving these little pheromones drops creating a data set that then can be handed over
to a data science team to analyze and hopefully come up with insights um yeah epic okay good luck ron

thank

you to all the live chatters thank you for presenting and looking forward to learning more thank you so much everyone bye bye bye

you